

5.4 Ratios and Proportions

Facebook Dominates the Social Media Landscape

Monthly active users of selected social networks and messaging services *

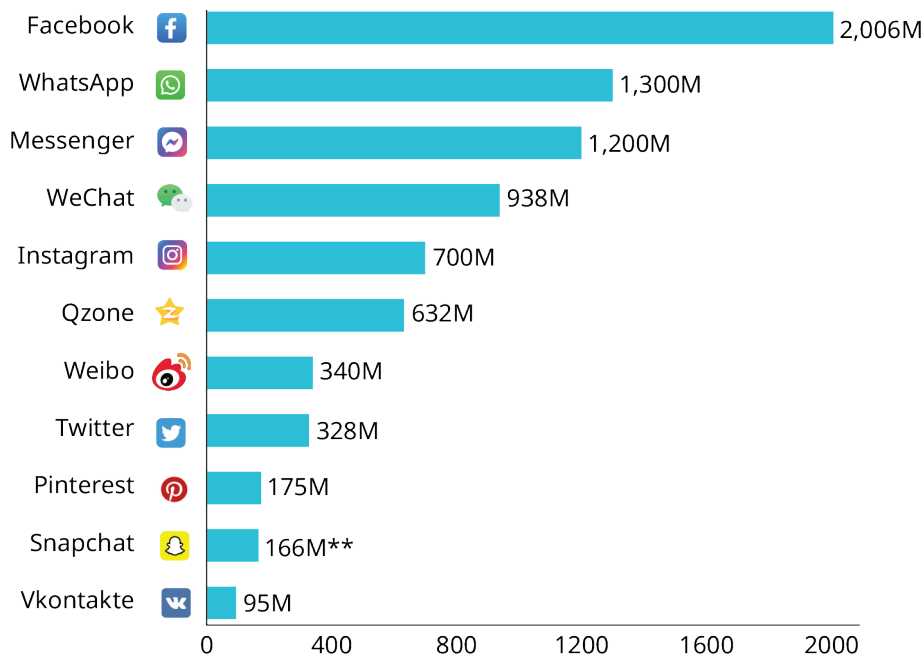


Figure 5.15 This bar graph shows popular social media app usage. (Source (https://openstax.org/r/media_apps_chart))

Learning Objectives

After completing this section, you should be able to:

1. Construct ratios to express comparison of two quantities.
2. Use and apply proportional relationships to solve problems.
3. Determine and apply a constant of proportionality.
4. Use proportions to solve scaling problems.

Ratios and proportions are used in a wide variety of situations to make comparisons. For example, using the information from [Figure 5.15](#), we can see that the number of Facebook users compared to the number of Twitter users is 2,006 M to 328 M. Note that the "M" stands for million, so 2,006 million is actually 2,006,000,000 and 328 million is 328,000,000. Similarly, the number of Qzone users compared to the number of Pinterest users is in a ratio of 632 million to 175 million. These types of comparisons are ratios.

Constructing Ratios to Express Comparison of Two Quantities

Note there are three different ways to write a **ratio**, which is a comparison of two numbers that can be written as: a to b OR $a : b$ OR the fraction a/b . Which method you use often depends upon the situation. For the most part, we will want to write our ratios using the fraction notation. Note that, while all ratios are fractions, not all fractions are ratios. Ratios make part to part, part to whole, and whole to part comparisons. Fractions make part to whole comparisons only.

EXAMPLE 5.28

Expressing the Relationship between Two Currencies as a Ratio

The Euro (€) is the most common currency used in Europe. Twenty-two nations, including Italy, France, Germany, Spain, Portugal, and the Netherlands use it. On June 9, 2021, 1 U.S. dollar was worth 0.82 Euros. Write this comparison as a ratio.

Solution

Using the definition of ratio, let $a = 1$ U.S. dollar and let $b = 0.82$ Euros. Then the ratio can be written as either 1 to 0.82; or 1:0.82; or $\frac{1}{0.82}$.

> YOUR TURN 5.28

1. On June 9, 2021, 1 U.S. dollar was worth 1.21 Canadian dollars. Write this comparison as a ratio.

EXAMPLE 5.29**Expressing the Relationship between Two Weights as a Ratio**

The gravitational pull on various planetary bodies in our solar system varies. Because weight is the force of gravity acting upon a mass, the weights of objects is different on various planetary bodies than they are on Earth. For example, a person who weighs 200 pounds on Earth would weigh only 33 pounds on the moon! Write this comparison as a ratio.

✓ Solution

Using the definition of ratio, let $a = 200$ pounds on Earth and let $b = 33$ pounds on the moon. Then the ratio can be written as either 200 to 33; or 200:33; or $\frac{200}{33}$.

> YOUR TURN 5.29

1. A person who weighs 170 pounds on Earth would weigh 64 pounds on Mars. Write this comparison as a ratio.

Using and Applying Proportional Relationships to Solve Problems

Using **proportions** to solve problems is a very useful method. It is usually used when you know three parts of the proportion, and one part is unknown. Proportions are often solved by setting up like ratios. If $\frac{a}{b}$ and $\frac{c}{d}$ are two ratios such that $\frac{a}{b} = \frac{c}{d}$, then the fractions are said to be **proportional**. Also, two fractions $\frac{a}{b}$ and $\frac{c}{d}$ are proportional ($\frac{a}{b} = \frac{c}{d}$) if and only if $a \times d = b \times c$.

EXAMPLE 5.30**Solving a Proportion Involving Two Currencies**

You are going to take a trip to France. You have \$520 U.S. dollars that you wish to convert to Euros. You know that 1 U.S. dollar is worth 0.82 Euros. How much money in Euros can you get in exchange for \$520?

✓ Solution

Step 1: Set up the two ratios into a proportion; let x be the variable that represents the unknown. Notice that U.S. dollar amounts are in both numerators and Euro amounts are in both denominators.

$$\frac{1}{0.82} = \frac{520}{x}$$

Step 2: Cross multiply, since the ratios $\frac{a}{b}$ and $\frac{c}{d}$ are proportional, then $a \times d = b \times c$.

$$\begin{aligned} 520(0.82) &= 1(x) \\ 426.4 &= x \end{aligned}$$

You should receive 426.4 Euros (426.4€).

> YOUR TURN 5.30

1. After your trip to France, you have 180 Euros remaining. You wish to convert them back into U.S. dollars. Assuming the exchange rate is the same ($\$1 = 0.82 \text{ €}$), how many dollars should you receive? Round to the nearest cent if necessary.

EXAMPLE 5.31**Solving a Proportion Involving Weights on Different Planets**

A person who weighs 170 pounds on Earth would weigh 64 pounds on Mars. How much would a typical racehorse (1,000 pounds) weigh on Mars? Round your answer to the nearest tenth.

 Solution

Step 1: Set up the two ratios into a proportion. Notice the Earth weights are both in the numerator and the Mars weights are both in the denominator.

$$\frac{170}{64} = \frac{1,000}{x}$$

Step 2: Cross multiply, and then divide to solve.

$$\begin{aligned} 170x &= 1,000(64) \\ 170x &= 64,000 \\ \frac{170x}{170} &= \frac{64,000}{170} \\ x &= 376.5 \end{aligned}$$

So the 1,000-pound horse would weigh about 376.5 pounds on Mars.

**YOUR TURN 5.31**

1. A person who weighs 200 pounds on Earth would weigh only 33 pounds on the moon. A 2021 Toyota Prius weighs 3,040 pounds on Earth; how much would it weigh on the moon? Round to the nearest tenth if necessary.

EXAMPLE 5.32**Solving a Proportion Involving Baking**

A cookie recipe needs $2\frac{1}{4}$ cups of flour to make 60 cookies. Jackie is baking cookies for a large fundraiser; she is told she needs to bake 1,020 cookies! How many cups of flour will she need?

 Solution

Step 1: Set up the two ratios into a proportion. Notice that the cups of flour are both in the numerator and the amounts of cookies are both in the denominator. To make the calculations more efficient, the cups of flour ($2\frac{1}{4}$) is converted to a decimal number (2.25).

$$\frac{2.25}{60} = \frac{x}{1020}$$

Step 2: Cross multiply, and then simplify to solve.

$$\begin{aligned} 2.25(1,020) &= 60x \\ 2,295 &= 60x \\ 38.25 &= x \end{aligned}$$

Jackie will need 38.25, or $38\frac{1}{4}$, cups of flour to bake 1,020 cookies.

**YOUR TURN 5.32**

1. You are going to bake cookies, using the same recipe as above. You find out that you have 27 cups of flour in your pantry. Assuming you have all the other ingredients necessary, how many cookies can you make with 27

cups of flour?

 Part of the definition of proportion states that two fractions $\frac{a}{b}$ and $\frac{c}{d}$ are proportional if $a \times d = b \times c$. This is the "cross multiplication" rule that students often use (and unfortunately, often use incorrectly). The only time cross multiplication can be used is if you have two ratios (and only two ratios) set up in a proportion. For example, you cannot use cross multiplication to solve for x in an equation such as $\frac{2}{5} = \frac{x}{8} + 3x$ because you do not have just the two ratios. Of course, you could use the rules of algebra to change it to be just two ratios and then you could use cross multiplication, but in its present form, cross multiplication cannot be used.



PEOPLE IN MATHEMATICS

Eudoxus was born around 408 BCE in Cnidus (now known as Knidos) in modern-day Turkey. As a young man, he traveled to Italy to study under Archytas, one of the followers of Pythagoras. He also traveled to Athens to hear lectures by Plato and to Egypt to study astronomy. He eventually founded a school and had many students.

Eudoxus made many contributions to the field of mathematics. In mathematics, he is probably best known for his work with the idea of proportions. He created a definition of proportions that allowed for the comparison of any numbers, even irrational ones. His definition concerning the equality of ratios was similar to the idea of cross multiplying that is used today. From his work on proportions, he devised what could be described as a method of integration, roughly 2000 years before calculus (which includes integration) would be fully developed by Isaac Newton and Gottfried Leibniz. Through this technique, Eudoxus became the first person to rigorously prove various theorems involving the volumes of certain objects. He also developed a planetary theory, made a sundial still usable today, and wrote a seven volume book on geography called *Tour of the Earth*, in which he wrote about all the civilizations on the Earth, and their political systems, that were known at the time. While this book has been lost to history, over 100 references to it by different ancient writers attest to its usefulness and popularity.

Determining and Applying a Constant of Proportionality

In the last example, we were given that $2\frac{1}{4}$ cups of flour could make 60 cookies; we then calculated that $38\frac{1}{4}$ cups of flour would make 1,020 cookies, and 720 cookies could be made from 27 cups of flour. Each of those three ratios is written as a fraction below (with the fractions converted to decimals). What happens if you divide the numerator by the denominator in each?

$$\frac{2.25}{60} = 0.0375 \quad \frac{38.25}{1,020} = 0.0375 \quad \frac{27}{720} = 0.0375.$$

The quotients in each are exactly the same! This number, determined from the ratio of cups of flour to cookies, is called the **constant of proportionality**. If the values a and b are related by the equality $\frac{a}{b} = k$, then k is the constant of proportionality between a and b . Note since $\frac{a}{b} = k$, then $b = \frac{a}{k}$, and $b = \frac{a}{k}$.

One piece of information that we can derive from the constant of proportionality is a unit rate. In our example (cups of flour divided by cookies), the constant of proportionality is telling us that it takes 0.0375 cups of flour to make one cookie. What if we had performed the calculation the other way (cookies divided by cups of flour)?

$$\frac{60}{2.25} = 26.6666... \quad \frac{1,020}{38.25} = 26.6666... \quad \frac{720}{27} = 26.6666...$$

In this case, the constant of proportionality ($26.6666... = 26\frac{2}{3}$) is telling us that $26\frac{2}{3}$ cookies can be made with one cup of flour. Notice in both cases, the "one" unit is associated with the denominator. The constant of proportionality is also useful in calculations if you only know one part of the ratio and wish to find the other.

EXAMPLE 5.33

Finding a Constant of Proportionality

Isabelle has a part-time job. She kept track of her pay and the number of hours she worked on four different days, and recorded it in the table below. What is the constant of proportionality, or pay divided by hours? What does the constant

of proportionality tell you in this situation?

Pay	\$87.50	\$50.00	\$37.50	\$100.00
Hours	7	4	3	8

 Solution

To find the constant of proportionality, divide the pay by hours using the information from any of the four columns. For example, $\frac{87.5}{7} = 12.5$. The constant of proportionality is 12.5, or \$12.50. This tells you Isabelle's hourly pay: For every hour she works, she gets paid \$12.50.

 YOUR TURN 5.33

- The following table contains the lengths of four objects in both inches and centimeters. What is the constant of proportionality (centimeters divided by inches)? What does the constant of proportionality tell you in this situation?

Object	floor tile	book	table	pencil
Length (in.)	24	13	60	7.5
Length (cm)	60.96	33.02	152.4	19.05

EXAMPLE 5.34

Applying a Constant of Proportionality: Running mph

Zac runs at a constant speed: 4 miles per hour (mph). One day, Zac left his house at exactly noon (12:00 PM) to begin running; when he returned, his clock said 4:30 PM. How many miles did he run?

 Solution

The constant of proportionality in this problem is 4 miles per hour (or 4 miles in 1 hour). Since $\frac{a}{b} = k$, where k is the constant of proportionality, we have

$$\frac{a \text{ miles}}{b \text{ hours}} = k$$

$$\frac{a}{4.5} = 4 \text{ (30 minutes is } \frac{1}{2}, \text{ or 0.5, hours)}$$

$$a = 4(4.5), \text{ since from the definition we know } a = kb$$

$$a = 18$$

Zac ran 18 miles.

 YOUR TURN 5.34

- One week, Zac ran a total of 122 miles. How much time did he spend running in that week?

EXAMPLE 5.35**Applying a Constant of Proportionality: Filling Buckets**

Joe had a job where every time he filled a bucket with dirt, he was paid \$2.50. One day Joe was paid \$337.50. How many buckets did he fill that day?

 Solution

The constant of proportionality in this situation is \$2.50 per bucket (or \$2.50 for 1 bucket). Since $\frac{a}{b} = k$, where k is the constant of proportionality, we have

$$\begin{aligned}\frac{a \text{ dollars}}{b \text{ buckets}} &= k \\ \frac{337.50}{b} &= 2.50\end{aligned}$$

Since we are solving for b , and we know from the definition that $b = \frac{a}{k}$:

$$\begin{aligned}b &= \frac{337.50}{2.50} \\ b &= 135\end{aligned}$$

Joe filled 135 buckets.

 YOUR TURN 5.35

1. Suppose one day Joe filled 83 buckets; how much money would he make on that day?

EXAMPLE 5.36**Applying a Constant of Proportionality: Miles vs. Kilometers**

While driving in Canada, Mabel quickly noticed the distances on the road signs were in kilometers, not miles. She knew the constant of proportionality for converting kilometers to miles was about 0.62—that is, there are about 0.62 miles in 1 kilometer. If the last road sign she saw stated that Montreal is 104 kilometers away, about how many more miles does Mabel have to drive? Round your answer to the nearest tenth.

 Solution

The constant of proportionality in this situation is 0.62 miles per 1 kilometer. Since $\frac{a}{b} = k$, where k is the constant of proportionality, we have

$$\begin{aligned}\frac{a \text{ miles}}{b \text{ kilometers}} &= k \\ \frac{a}{104} &= 0.62 \\ a &= 0.62(104) \\ a &= 64.48\end{aligned}$$

Rounding the answer to the nearest tenth, Mabel has to drive 64.5 miles.

 YOUR TURN 5.36

1. Later in her trip, Mabel decides to drive to the capital of Canada, Ottawa. As she left Montreal, she saw a road sign that read that Ottawa is 203 kilometers away. About how many miles is that? Round your answer to the nearest tenth.

Using Proportions to Solve Scaling Problems



Figure 5.16 A map of the northeastern United States

Ratio and proportions are used to solve problems involving **scale**. One common place you see a scale is on a map (as represented in [Figure 5.16](#)). In this image, 1 inch is equal to 200 miles. This is the scale. This means that 1 inch on the map corresponds to 200 miles on the surface of Earth. Another place where scales are used is with models: model cars, trucks, airplanes, trains, and so on. A common ratio given for model cars is 1:24—that means that 1 inch in length on the model car is equal to 24 inches (2 feet) on an actual automobile. Although these are two common places that scale is used, it is used in a variety of other ways as well.

EXAMPLE 5.37

Solving a Scaling Problem Involving Maps

[Figure 5.17](#) is an outline map of the state of Colorado and its counties. If the distance of the southern border is 380 miles, determine the scale (i.e., 1 inch = how many miles). Then use that scale to determine the approximate lengths of the other borders of the state of Colorado.

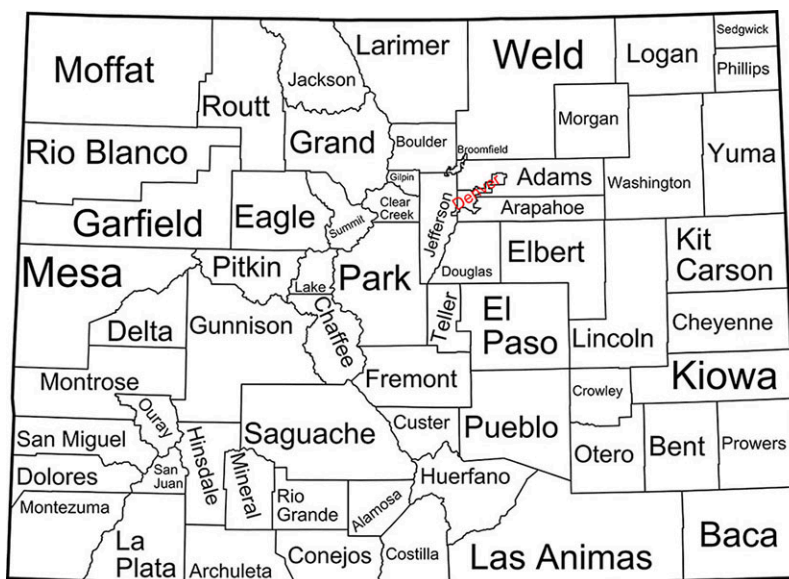


Figure 5.17 Outline Map of Colorado (credit: "Map of Colorado Counties" by David Benbennick/Wikimedia Commons,

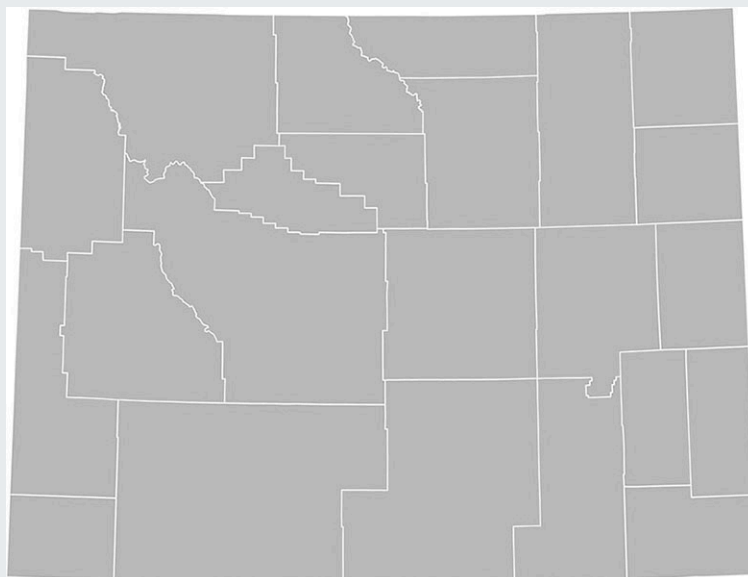
Public Domain)

✓ **Solution**

When the southern border is measured with a ruler, the length is 4 inches. Since the length of the border in real life is 380 miles, our scale is 1 inch = 95 miles.

The eastern and western borders both measure 3 inches, so their lengths are about 285 miles. The northern border measures the same as the southern border, so it has a length of 380 miles.

> **YOUR TURN 5.37**



1.

Outline Map of Wyoming (credit: "Blank map of Wyoming showing counties" by David Benbennick/Wikimedia Commons, Public Domain)

Consider the outline map of the state of Wyoming and its counties. If the distance of the southern border is 365 miles, determine the scale (i.e., 1 inch = how many miles). Then use that scale to determine the approximate lengths of the other borders of the state of Wyoming.

EXAMPLE 5.38

Solving a Scaling Problem Involving Model Cars

Die-cast NASCAR model cars are said to be built on a scale of 1:24 when compared to the actual car. If a model car is 9 inches long, how long is a real NASCAR automobile? Write your answer in feet.

✓ **Solution**

The scale tells us that 1 inch of the model car is equal to 24 inches (2 feet) on the real automobile. So set up the two ratios into a proportion. Notice that the model lengths are both in the numerator and the NASCAR automobile lengths are both in the denominator.

$$\begin{aligned}\frac{1}{24} &= \frac{9}{x} \\ 24(9) &= x \\ 216 &= x\end{aligned}$$

This amount (216) is in inches. To convert to feet, divide by 12, because there are 12 inches in a foot (this conversion from inches to feet is really another proportion!). The final answer is:

$$\frac{216}{12} = 18$$

The NASCAR automobile is 18 feet long.

> YOUR TURN 5.38

1. A toy Jeep is built on a 1 : 16 scale. The website for the toy Jeep says the toy is 11.5 inches long. Based on this, how long is the real Jeep?

Check Your Understanding

32. If $a:b = c:d$, then $b:a = d:c$ for all non-zero whole numbers a , b , c , and d .
 - a. True
 - b. False
33. If the ratio of wolves to rabbits in a national park is 3:5, then the ratio of rabbits to (wolves and rabbits) is 5:8.
 - a. True
 - b. False
34. All fractions are ratios but not all ratios are fractions.
 - a. True
 - b. False
35. In the following equation, $\frac{x}{46} = \frac{20}{90} + 4$, cross multiplication can be used as the first step towards solving for x .
 - a. True
 - b. False
36. All fractions are ratios but not all ratios are fractions.
 - a. True
 - b. False
37. There are 16 math majors and 12 non-math majors in Ms. Kraft's class. What is **not** a correct way to express the ratio of math majors to non-math majors?

16 : 12	12 : 16	4 : 3
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38. There are 16 math majors and 12 non-math majors in Ms. Kraft's class. What shows the ratio of math majors to all the students in Ms. Kraft's class?

16 : 12	12 : 16	16 : 28	28 : 12	None of these
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39. One U.S. dollar is worth 0.72 British pounds. Damon is traveling to Great Britain and wishes to exchange \$450 U.S. dollars for British pounds. How many British pounds should Damon get in return?

625	6,250	3,456	345.6	None of these
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40. The HO scale for model trains is the most common size of model trains. This scale is 1 : 87. If a real locomotive is 73 feet long, how long should the model locomotive be (in inches)? Round your answer to the nearest inch.
41. Albert's Honda Civic gets 37 miles per gallon of gasoline. The gas tank on the Civic can hold 13.5 gallons of gas. Albert is driving from Tucson, Arizona to Los Angeles, California, a distance of 485 miles. Albert thinks he can make it on one full tank of gasoline. Can he? Explain.

42. The average price of a gallon of regular gasoline in the California on July 1, 2021 was \$4.28 per gallon. Albert stops at a gas station in California and puts 9.5 gallons of gasoline into his Civic. How much did he pay for the gas?



SECTION 5.4 EXERCISES

For the following exercises, use this scenario: Kelly opened a bag of colored chocolate coated candies and counted the number of each color of candy. She found she had 9 green, 4 yellow, 13 black, 11 orange, 8 blue, and 7 red. What is the ratio of the following candy colors?

1. Red candies to green candies
2. Green candies to black candies
3. Yellow candies to black candies
4. Black candies to blue candies
5. Orange candies to non-orange candies
6. Yellow candies to non-yellow candies
7. Red candies to all candies
8. Pink candies to all candies
9. Candies with the letter 'r' in their name to all candies
10. Candies with the letter 'r' in their name to candies without the letter 'r' in their name

For the following exercises, solve each proportion for the unknown variable.

11. $\frac{x}{46} = \frac{4}{8}$
12. $\frac{27}{x} = \frac{3.5}{14}$
13. $\frac{16}{7} = \frac{x}{14}$
14. $\frac{1}{0.8} = \frac{x}{514}$
15. $\frac{203}{g} = \frac{10}{22}$
16. $\frac{1}{29.5} = \frac{16}{m}$
17. $\frac{14}{95} = \frac{a}{19}$
18. $\frac{13}{140} = \frac{s}{2961}$
19. $\frac{1}{k} = \frac{69}{111}$ (Round answer to the nearest hundredth.)
20. $\frac{p}{1} = \frac{22}{7}$ (Round answer to the nearest hundredth.)
21. Pet Paradise has 20 cats and 16 dogs. Animal Acres has 15 cats. How many dogs must be at Animal Acres so that Pet Paradise and Animal Acres have the same ratio of cats to dogs?
22. Pet Paradise has 20 cats and 16 dogs. Critter Corral has 28 dogs. How many cats must be at Critter Corral so that Pet Paradise and Critter Corral have the same ratio of cats to dogs?
23. A high school has 960 students. The ratio of students to high school teachers is 16 : 1. How many high school teachers are at the school?
24. A high school has 960 students. The ratio of students to high school teachers is 16 : 1. How many more teachers are needed to have a 12 : 1 ratio at the high school of students to teachers?
25. One U.S. dollar is worth \$1.23 Canadian dollars. Bernice is traveling to Canada and wants to convert \$550 U.S. to Canadian money. How much in Canadian money should she receive?
26. One U.S. dollar is worth \$1.23 Canadian dollars. Rene is traveling from Canada to the United States and wants to convert \$550 of Canadian money to U.S. money. How much in U.S. money should he receive? Round your answer to the nearest cent.
27. One U.S. dollar is worth \$1.23 Canadian dollars. What is one Canadian dollar worth in U.S. funds? Round your answer to the nearest cent.
28. A salad recipe needs one cup of crushed almonds. It will serve eight people. Rashida needs to make a salad to serve 20 people. How many cups of crushed almonds does she need?
29. A salad recipe needs one cup of crushed almonds. It will serve eight people. Elmer has 4.75 cups of crushed almonds. If he uses all of the crushed almonds he has to make this salad, how many people will it serve?
30. Jorge is 6 feet tall and casts a 7-foot shadow. At the same time, a nearby tree has a shadow of 56 feet. How tall is the tree?